

Volume I Foundational Geometry Reference

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Euclidean Geometry

Definitions

Definition 0.0.1 (Point). A point is that which has no part.

Definition 0.0.2 (Line). A line is breadthless length.

Definition 0.0.3 (Ends of a line). The ends of a line are points.

Definition 0.0.4 (Straight line). A straight line is a line which lies evenly with the points on itself.

Definition 0.0.5 (Surface). A surface is that which has length and breadth only.

Definition 0.0.6 (Edges of a surface). The edges of a surface are lines.

Definition 0.0.7 (Plane surface). A plane surface is a surface which lies evenly with the straight lines on itself.

Definition 0.0.8 (Plane angle). A plane angle is the inclination to one another of two lines in a plane which meet one another and do not lie in a straight line.

Definition 0.0.9 (Rectilinear angle). And when the lines containing the angle are straight, the angle is called rectilinear.

Definition 0.0.10 (Right angle and perpendicular). When a straight line standing on a straight line makes the adjacent angles equal to one another, each of the equal angles is right, and the straight line standing on the other is called a perpendicular to that on which it stands.

Definition 0.0.11 (Obtuse angle). An obtuse angle is an angle greater than a right angle.

Definition 0.0.12 (Acute angle). An acute angle is an angle less than a right angle.

Definition 0.0.13 (Boundary). A boundary is that which is an extremity of anything.

Definition 0.0.14 (Figure). A figure is that which is contained by any boundary or boundaries.

Definition 0.0.15 (Circle). A circle is a plane figure contained by one line such that all the straight lines falling upon it from one point among those lying within the figure equal one another.

Definition 0.0.16 (Center of a circle). And the point is called the center of the circle.

Definition 0.0.17 (Diameter). A diameter of the circle is any straight line drawn through the center and terminated in both directions by the circumference of the circle, and such a straight line also bisects the circle.

Definition 0.0.18 (Semicircle). A semicircle is the figure contained by the diameter and the circumference cut off by it. And the center of the semicircle is the same as that of the circle.

Definition 0.0.19 (Rectilinear figures). Rectilinear figures are those which are contained by straight lines, trilateral figures being those contained by three, quadrilateral those contained by four, and multilateral those contained by more than four straight lines.

Definition 0.0.20 (Equilateral, isosceles, and scalene triangles). Of trilateral figures, an equilateral triangle is that which has its three sides equal, an isosceles triangle that which has two of its sides alone equal, and a scalene triangle that which has its three sides unequal.

Definition 0.0.21 (Right-angled, obtuse-angled, and acute-angled triangles). Further, of trilateral figures, a right-angled triangle is that which has a right angle, an obtuse-angled triangle that which has an obtuse angle, and an acute-angled triangle that which has its three angles acute.

Definition 0.0.22 (Quadrilateral classes). Of quadrilateral figures, a square is that which is both equilateral and right-angled; an oblong that which is right-angled but not equilateral; a rhombus that which is equilateral but not right-angled; and a rhomboid that which has its opposite sides and angles equal to one another but is neither equilateral nor right-angled. And let quadrilaterals other than these be called trapezia.

Definition 0.0.23 (Parallel straight lines). Parallel straight lines are straight lines which, being in the same plane and being produced indefinitely in both directions, do not meet one another in either direction.

Postulates

0.1 Foundations

Axioms

Propositions

Theorem 0.1.1 (Euclid I.1: Constructing an equilateral triangle). *Given a finite straight line segment AB , there exists an equilateral triangle ABC constructed on AB .*

Theorem 0.1.2 (Euclid I.2: Copying a segment from a given point). *Given a point A and a finite straight line segment BC , there exists a straight line segment beginning at A and equal to BC .*

Theorem 0.1.3 (Euclid I.3: Cutting off an equal segment). *Given two unequal finite straight line segments, there exists a construction which cuts off from the greater a straight line segment equal to the less.*

Theorem 0.1.4 (Euclid I.4: Side-angle-side congruence). *If two triangles have two sides equal to two sides respectively, and have the angles contained by the equal straight lines equal, then they also have the base equal to the base, the triangle equal to the triangle, and the remaining angles equal to the remaining angles respectively.*

Theorem 0.1.5 (Euclid I.31: Drawing a parallel through a point). *Through a given point, to draw a straight line parallel to a given straight line.*

Theorem 0.1.6 (Euclid I.32: Triangle angle sum). *In any triangle, if one of the sides is produced, then the exterior angle is equal to the two interior and opposite angles, and the three interior angles of the triangle are equal to two right angles.*

Theorem 0.1.7 (Euclid I.41: Triangle and parallelogram on the same base). *If a parallelogram has the same base with a triangle and is in the same parallels, then the parallelogram is double the triangle.*

Theorem 0.1.8 (Euclid I.46: Constructing a square on a segment). *On a given straight line, to describe a square.*

Theorem 0.1.9 (Euclid I.47: Pythagorean theorem). *In right-angled triangles, the square on the side subtending the right angle is equal to the squares on the sides containing the right angle.*

0.2 Compass-and-Straightedge Constructions

Proofs

Trigonometry

0.2.1 Trigonometric Functions from Euclidean Geometry

Definition 0.2.1 (Right-triangle sine). Let T be a right triangle and let θ be one of its acute angles. Write

$$\text{opp}(T, \theta), \quad \text{adj}(T, \theta), \quad \text{hyp}(T)$$

for the side lengths opposite θ , adjacent to θ , and opposite the right angle, respectively. Define

$$\sin \theta = \frac{\text{opp}(T, \theta)}{\text{hyp}(T)}.$$

Definition 0.2.2 (Right-triangle cosine). Let T be a right triangle and let θ be one of its acute angles. Define

$$\cos \theta = \frac{\text{adj}(T, \theta)}{\text{hyp}(T)}.$$

Definition 0.2.3 (Right-triangle tangent). Let T be a right triangle and let θ be one of its acute angles. Define

$$\tan \theta = \frac{\text{opp}(T, \theta)}{\text{adj}(T, \theta)}.$$

Definition 0.2.4 (Right-triangle cotangent). Let T be a right triangle and let θ be one of its acute angles. Define

$$\cot \theta = \frac{\text{adj}(T, \theta)}{\text{opp}(T, \theta)}.$$

Definition 0.2.5 (Right-triangle secant). Let T be a right triangle and let θ be one of its acute angles. Define

$$\sec \theta = \frac{\text{hyp}(T)}{\text{adj}(T, \theta)}.$$

Definition 0.2.6 (Right-triangle cosecant). Let T be a right triangle and let θ be one of its acute angles. Define

$$\csc \theta = \frac{\text{hyp}(T)}{\text{opp}(T, \theta)}.$$

Definition 0.2.7 (Acute-angle sine). For an acute Euclidean angle θ , choose any right triangle T having θ as an acute angle and set

$$\sin \theta = \frac{\text{opp}(T, \theta)}{\text{hyp}(T)}.$$

Definition 0.2.8 (Acute-angle cosine). For an acute Euclidean angle θ , choose any right triangle T having θ as an acute angle and set

$$\cos \theta = \frac{\text{adj}(T, \theta)}{\text{hyp}(T)}.$$

Theorem 0.2.9 (Similarity invariance of trigonometric ratios). *If two right triangles have the same acute angle θ , then their corresponding opposite-to-hypotenuse and adjacent-to-hypotenuse ratios are equal.*

Theorem 0.2.10 (Well-definedness of sine and cosine). *For each acute Euclidean angle θ , the values $\sin \theta$ and $\cos \theta$ are independent of the right triangle chosen to compute them.*

Theorem 0.2.11 (Six trigonometric functions). *The six right-triangle trigonometric ratios define functions of an acute Euclidean angle, subject to the denominator restrictions in their defining quotients.*

Definition 0.2.12 (Unit-circle cosine). Let θ be an angle in standard position, and let $P(\theta) = (x_\theta, y_\theta)$ be the point where its terminal ray meets the unit circle. Define

$$\cos \theta = x_\theta.$$

Definition 0.2.13 (Unit-circle sine). Let θ be an angle in standard position, and let $P(\theta) = (x_\theta, y_\theta)$ be the point where its terminal ray meets the unit circle. Define

$$\sin \theta = y_\theta.$$

Theorem 0.2.14 (Unit-circle agreement with right-triangle trigonometry). *For $0 < \theta < 90^\circ$, the unit-circle definitions of sine and cosine agree with the right-triangle definitions.*

Definition 0.2.15 (Extended tangent). Where $\cos \theta \neq 0$, define

$$\tan \theta = \frac{\sin \theta}{\cos \theta}.$$

Definition 0.2.16 (Extended cotangent). Where $\sin \theta \neq 0$, define

$$\cot \theta = \frac{\cos \theta}{\sin \theta}.$$

Definition 0.2.17 (Extended secant). Where $\cos \theta \neq 0$, define

$$\sec \theta = \frac{1}{\cos \theta}.$$

Definition 0.2.18 (Extended cosecant). Where $\sin \theta \neq 0$, define

$$\csc \theta = \frac{1}{\sin \theta}.$$

Definition 0.2.19 (Radian measure). One radian is the angle that cuts off arc length 1 on the unit circle. A full turn has measure 2π radians.

Theorem 0.2.20 (Symmetry and periodicity of sine and cosine). *For every angle in the unit-circle domain,*

$$\cos(-\theta) = \cos \theta, \quad \sin(-\theta) = -\sin \theta,$$

and one full turn returns the same sine and cosine values.

Theorem 0.2.21 (Pythagorean identity). *For every angle in the unit-circle domain,*

$$\cos^2 \theta + \sin^2 \theta = 1.$$

Theorem 0.2.22 (Reciprocal identities). *Where both sides are defined,*

$$\sec \theta = \frac{1}{\cos \theta}, \quad \csc \theta = \frac{1}{\sin \theta}, \quad \cot \theta = \frac{1}{\tan \theta}.$$

Theorem 0.2.23 (Quotient identities). *Where both sides are defined,*

$$\tan \theta = \frac{\sin \theta}{\cos \theta}, \quad \cot \theta = \frac{\cos \theta}{\sin \theta}.$$

Theorem 0.2.24 (Tangent and secant Pythagorean identities). *Where both sides are defined,*

$$1 + \tan^2 \theta = \sec^2 \theta, \quad 1 + \cot^2 \theta = \csc^2 \theta.$$

Theorem 0.2.25 (Angle-sum identities). *For angles in the unit-circle domain,*

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta,$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta.$$

Theorem 0.2.26 (Angle-difference identities). *For angles in the unit-circle domain,*

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta,$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta.$$

Theorem 0.2.27 (Double-angle identities). *For angles in the unit-circle domain,*

$$\sin(2\theta) = 2 \sin \theta \cos \theta,$$

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta.$$

Theorem 0.2.28 (Half-angle identities). *Where the signs are chosen according to the quadrant of $\theta/2$,*

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}, \quad \cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}.$$

Theorem 0.2.29 (Product-to-sum identities). *For angles in the unit-circle domain,*

$$\sin \alpha \cos \beta = \frac{\sin(\alpha + \beta) + \sin(\alpha - \beta)}{2},$$

$$\cos \alpha \cos \beta = \frac{\cos(\alpha + \beta) + \cos(\alpha - \beta)}{2},$$

$$\sin \alpha \sin \beta = \frac{\cos(\alpha - \beta) - \cos(\alpha + \beta)}{2}.$$

Theorem 0.2.30 (Sum-to-product identities). *For angles in the unit-circle domain,*

$$\sin u + \sin v = 2 \sin \frac{u+v}{2} \cos \frac{u-v}{2},$$

$$\cos u + \cos v = 2 \cos \frac{u+v}{2} \cos \frac{u-v}{2},$$

$$\cos u - \cos v = -2 \sin \frac{u+v}{2} \sin \frac{u-v}{2}.$$

Proofs

Analytical Geometry

0.3 Analytical Geometry

0.4 Lines

Definition 0.4.1 (Point-slope form). Let $P_1 = (x_1, y_1)$ be a point in the Euclidean plane and let $m \in \mathbb{R}$. The point-slope form of the line through P_1 with slope m is

$$y - y_1 = m(x - x_1).$$

Definition 0.4.2 (Two-point form). Let $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ be distinct points with $x_1 \neq x_2$. The two-point form of the line through P_1 and P_2 is

$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (x - x_1).$$

Definition 0.4.3 (General linear form). The general linear form of a line in $\mathbb{R}^2$ is

$$Ax + By + C = 0,$$

where $A, B, C \in \mathbb{R}$ and $A^2 + B^2 \neq 0$.

Definition 0.4.4 (Vector parametric form). A line in $\mathbb{R}^2$ may be represented as the image of $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^2$ given by

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v},$$

where $\mathbf{r}_0 \in \mathbb{R}^2$ is a point on the line and $\mathbf{v} \in \mathbb{R}^2$ is a nonzero direction vector.

Definition 0.4.5 (Intercept form). If a line meets the x -axis at $(a, 0)$ and the y -axis at $(0, b)$ with $a, b \neq 0$, its intercept form is

$$\frac{x}{a} + \frac{y}{b} = 1.$$

Definition 0.4.6 (Affine parametric line in $\mathbb{R}^n$). Given distinct points $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$, the line through them is

$$\mathbf{x}(t) = \mathbf{x}_1 + t(\mathbf{x}_2 - \mathbf{x}_1), \quad t \in \mathbb{R}.$$

Definition 0.4.7 (Normal form). A line in $\mathbb{R}^2$ may be written in normal form as

$$x \cos \alpha + y \sin \alpha - p = 0,$$

where $p \geq 0$ is the distance from the origin to the line.

Definition 0.4.8 (Symmetric form). Given a point (x_1, y_1) and direction cosines $(\cos \theta, \sin \theta)$, the symmetric form of a line is

$$\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r.$$

Proofs